

Measure Theory

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0.1 Measure

It will be inevitable to step on some measure & integration theory in dealing with *Partial Differential Equations* and *Stochastic Processes*. This chapter acts as a brief introduction.

0.1.1 σ -Algebra

Motivation

In physics, the task of finding “volume” is ubiquitous. For instance in computing mass, charge, the integral would resemble:

$$\int (\cdot) dV \quad (1)$$

In some sense, this dV is a function $dV : E \rightarrow \mathbb{R}$ which computes the volume of $E \subset \mathbb{R}^n$, a tiny control volume we are interested in. It would be very natural to ask for this function to satisfy:

Definition 0.1.1 (Measure). Suppose X is a set. Let $\mu : A \rightarrow \mathbb{R}$ where $A \in P(X)$ such that:

•

$$\mu(\emptyset) = 0 \quad (2)$$

• (**countable additivity**) $E_1, E_2, \dots$ is a countable infinite sequence of disjoint, then:

$$\mu(\cup_j E_j) = \sum_j \mu(E_j) \quad (3)$$

Then we call μ a measure.

The question would be, when can we find such a function. Trivially we can obtain such property by sending every A to 0. However, for more practical purposes of modeling the world, we would like nicer properties. In particular, let's consider this completely reasonable demand on $\mathbb{R}^n$.

Exercise 1 (Open Set as a Topology)

1. Let $\mathcal{B}$ be the family of all open intervals in $\mathbb{R}$, i.e. subsets of the form

$$(a, b) := \{c \in \mathbb{R} \mid a < c < b\} \quad (4)$$

for all $a, b \in \mathbb{R}$ with $a < b$. Let $\mathcal{T}(\mathcal{B})$ be the family of all arbitrary unions of elements in $\mathcal{B}$, i.e.,

$$\mathcal{T}(\mathcal{B}) := \left\{ \bigcup_{B \in \mathcal{B}'} B \mid \mathcal{B}' \subseteq \mathcal{B} \right\} \quad (5)$$

Prove that $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$. (This “ $\mathcal{B}$ ” stands for a “base” = “basis” of a topology.)

2. For $\mathcal{B}$ and $\mathcal{T}(\mathcal{B})$ as above and any $U \in \mathcal{T}(\mathcal{B})$ (i.e., any open set), prove that there exists a countable family $\mathcal{B}'' \subseteq \mathcal{B}$ (of open intervals) such that $U = \bigcup_{B \in \mathcal{B}''} B$. (Hint: it might help to prove first that the union of any family of open intervals containing a given $x \in \mathbb{R}$ is a generalized open interval in the sense that it is of the form (a, b) , $(-\infty, a)$ or (a, ∞) for some $a, b \in \mathbb{R}$. Or do this any other way.)
3. For $\mathcal{T}(\mathcal{B})$ as above, prove that $\mathcal{T}(\mathcal{B}) \subseteq \mathcal{B}_{\mathbb{R}}$, the Borel σ -algebra on $\mathbb{R}$. (This “ $\mathcal{B}$ ”, confusingly, stands for “Borel”.)
4. Use the above to give a full proof of Proposition 1.2 on p. 22 in [F].
5. Also deduce that $\mathcal{M}(\mathcal{B}) = \mathcal{M}(\mathcal{T}(\mathcal{B})) = \mathcal{B}_{\mathbb{R}}$.

Solution to Exercise 1

1. We prove that $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$.

• **Property 1:** $\emptyset, \mathbb{R} \in \mathcal{T}(\mathcal{B})$

- $\emptyset = \cup_{B \in \emptyset} B$ (empty union), so $\emptyset \in \mathcal{T}(\mathcal{B})$
- $\mathbb{R} = (-\infty, \infty)$ is a generalized open interval, hence $\mathbb{R} \in \mathcal{T}(\mathcal{B})$. More explicitly, we can cover it by $\cup_{a=0}^{\infty} \{(a, a+1), (-a-1, -a)\}$

• **Property 2: Closed under arbitrary unions**

- Let $\mathcal{U} \subseteq \mathcal{T}(\mathcal{B})$ be an arbitrary collection of sets in $\mathcal{T}(\mathcal{B})$. For each $U \in \mathcal{U}$, we have $U = \cup_{B \in \mathcal{B}_U} B$ for some $\mathcal{B}_U \subseteq \mathcal{B}$. Then $\cup_{U \in \mathcal{U}} U = \cup_{U \in \mathcal{U}} (\cup_{B \in \mathcal{B}_U} B) = \cup_{B \in \cup_{U \in \mathcal{U}} \mathcal{B}_U} B$. Since $\cup_{U \in \mathcal{U}} \mathcal{B}_U \subseteq \mathcal{B}$, we have $\cup_{U \in \mathcal{U}} U \in \mathcal{T}(\mathcal{B})$

• **Property 3: Closed under finite intersections**

- By induction on the number n of sets.
- **Base case** ($n = 1$): Trivial, $U \in \mathcal{T}(\mathcal{B})$.
- **Inductive step:** Assume $U_1 \cap \cdots \cap U_n \in \mathcal{T}(\mathcal{B})$ for any n sets in $\mathcal{T}(\mathcal{B})$. For $n + 1$ sets, write $(U_1 \cap \cdots \cap U_n) \cap U_{n+1}$. By induction hypothesis, $U_1 \cap \cdots \cap U_n = \cup_{i \in I} (a_i, b_i)$ for some countable collection of open intervals. Then

$$(U_1 \cap \cdots \cap U_n) \cap U_{n+1} = (\cup_{i \in I} (a_i, b_i)) \cap U_{n+1} = \cup_{i \in I} ((a_i, b_i) \cap U_{n+1}) \quad (6)$$

Each $(a_i, b_i) \cap U_{n+1}$ is an intersection of an open interval with a union of open intervals, which is a finite union of open intervals (or empty). Hence $(U_1 \cap \cdots \cap U_{n+1}) \in \mathcal{T}(\mathcal{B})$.

Therefore $\mathcal{T}(\mathcal{B})$ is a topology on $\mathbb{R}$.

2. Suppose $\mathcal{B}' \subseteq \mathcal{B}$ and

$$U = \bigcup_{B \in \mathcal{B}'} B \quad (7)$$

where $\mathcal{B}'$ can be uncountable. We would like to show $\exists \mathcal{B}''$ that is countable such that

$$U = \bigcup_{B \in \mathcal{B}''} B \quad (8)$$

We now show a specific way to construct $\mathcal{B}''$ from $\mathcal{B}'$. Define:

$$\mathcal{B}_x = \{(a, b) \in \mathcal{B}' : x \in (a, b)\} \quad (9)$$

Clearly $\mathcal{B}_x \subseteq \mathcal{B}'$. Now proceed with the following procedure (an “infinite algorithm”) on the construction of $\mathcal{B}''$

Algorithm 0.1.1. Input: A potentially uncountable set of open intervals $\mathcal{B}'$ such that $\cup \mathcal{B}' = U$
Output: A countable set of open intervals $\mathcal{B}''$ such that $\cup \mathcal{B}'' = U$

1. $\mathcal{B}'' = \emptyset$

2. **While** $\exists x \in U, x \notin \cup \mathcal{B}''$:

(a) Compute $(c, d) = \cup_{B \in \mathcal{B}_x} B$ where c, d is in extended real number system.

(b) **If** $c, d \in \mathbb{R}$:

- i. $\mathcal{B}'_x = \{(c, d)\}$
- (c) **Else if** $c \in \mathbb{R}, d = \infty$:
 - i. $\mathcal{B}'_x \leftarrow \{(c, [c])\} \cup \{[c], [c] + j\}_{j=1}^\infty$
- (d) **Else** Other cases follows as (3)
- (e) $\mathcal{B}'' = \mathcal{B}'' \cup \mathcal{B}'_x$

3. Return $\mathcal{B}''$

where $[c]$ picks the closes integer towards the right of the real axis. Note we implicitly invoked the claim below in computing (c, d) :

- **Claim 1** The union of arbitrary family of open intervals containing $x \in \mathbb{R}$ is a *generalized open interval*

$$\bigcup_{B \in \mathcal{B}_x} B \in \{(c, d) : c, d \in \mathbb{R} \cup \{\infty, -\infty\}\} \quad (10)$$

- I claim the union is just (c, d) where $c = \inf\{a : (a, b) \in \mathcal{B}_x\}$ and $d = \sup\{b : (a, b) \in \mathcal{B}_x\}$ where we allow ∞ and $-\infty$. Clearly (c, d) covers $\bigcup \mathcal{B}_x$. It suffice to show $\bigcup \mathcal{B}_x$ covers (c, d) . Suppose $\exists y \in (c, d)$ and $y \notin \bigcup \mathcal{B}_x$. This means $\bigcup \mathcal{B}_x$ has a hole and there must exists a pair of open interval in $\mathcal{B}_x$ that is disjoint, contradicting that all open intervals in $\mathcal{B}_x$ contains x (in that case no pair should be disjoint).

Now what's left to show is (1) Only countable number of $\mathcal{B}'_x$ is needed. (2) each $\mathcal{B}'_x$ is countable. The later is obvious by construction. The former can be argued by countability of $\mathbb{Q}$. Pick a $q \in \mathbb{Q}$ in one of the open interval of $\mathcal{B}'_x$, this is always possible by density of rationals. Suppose uncountable $\mathcal{B}'_x$ is needed to cover $\bigcup \mathcal{B}''$ then $U = \bigcup \mathcal{B}''$ would contain uncountable number of rationals. This finishes the proof.

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3. By definition, $\mathcal{B}_{\mathbb{R}}$ is the smallest σ -algebra containing $\mathcal{T}(\mathcal{B})$. Therefore the inclusion holds by construction.
-

4. We only need to show (b), (c), and (e) since (a) and (d) are generalized open intervals and are actually shown by (5).

Case (b): Closed intervals $\mathcal{E}_2 = \{[a, b] : a < b\}$

- $\mathcal{M}(\mathcal{E}_2) \subseteq \mathcal{B}_{\mathbb{R}}$: Note that $[a, b] = \bigcap_{j=1}^\infty (a - 1/j, b + 1/j)$ is a countable intersection of open intervals, so $\mathcal{E}_2 \subset \mathcal{B}_{\mathbb{R}}$.
- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_2)$: $(a, b) = \bigcup_{j=1}^\infty [a + 1/j, b - 1/j]$ is a countable union of closed intervals, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_2)$ and hence $\mathcal{B}_{\mathbb{R}} = \mathcal{M}(\mathcal{B}) \subset \mathcal{M}(\mathcal{E}_2)$.

Case (c): Half-open intervals $\mathcal{E}_3 = \{(a, b] : a < b\}$

- $\mathcal{M}(\mathcal{E}_3) \subseteq \mathcal{B}_{\mathbb{R}}$: $(a, b] = (a, b) \cup \{b\}$ where $(a, b) \in \mathcal{B}_{\mathbb{R}}$ and $\{b\} = \bigcap_{j=1}^\infty (b - 1/j, b + 1/j) \in \mathcal{B}_{\mathbb{R}}$.
- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_3)$: $(a, b) = \bigcup_{j=1}^\infty (a, b - 1/j]$ is a countable union of half-open intervals, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_3)$ and hence $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_3)$.

Case (e): Closed rays $\mathcal{E}_7 = \{[a, \infty) : a \in \mathbb{R}\}$

- $\mathcal{M}(\mathcal{E}_7) \subseteq \mathcal{B}_{\mathbb{R}}$: $[a, \infty) = \bigcap_{j=1}^\infty (a - 1/j, \infty)$ is a countable intersection of open rays (generalized open intervals), so $\mathcal{E}_7 \subset \mathcal{B}_{\mathbb{R}}$.

- $\mathcal{B}_{\mathbb{R}} \subseteq \mathcal{M}(\mathcal{E}_7)$: $(a, b) = (a, \infty) \cap (-\infty, b)$ where $(a, \infty) = \bigcup_{j=1}^{\infty} [a + 1/j, \infty)$ and $(-\infty, b) = \mathbb{R} \setminus [b, \infty)$ are in $\mathcal{M}(\mathcal{E}_7)$, so $\mathcal{B} \subset \mathcal{M}(\mathcal{E}_7)$ and hence $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}(\mathcal{E}_7)$.

Cases (c'), (e') follow by symmetry.

5. We only need to show

$$\begin{aligned} \mathcal{B} &\subseteq \mathcal{M}(\mathcal{T}(\mathcal{B})) \\ \mathcal{T}(\mathcal{B}) &\subseteq \mathcal{M}(\mathcal{B}) \end{aligned} \tag{11}$$

- The former is trivial since $\mathcal{B} \subseteq \mathcal{T}(\mathcal{B}) \subseteq \mathcal{M}(\mathcal{T}(\mathcal{B}))$
- The later can be shown using (2). $\mathcal{M}(\mathcal{B})$ contains all countable unions of open intervals $\mathcal{B}$ while $\mathcal{T}(\mathcal{B})$ contains arbitrary union. (2) bridges the gap by claiming the arbitrary union can actually be replaced by countable union.

0.2 Construction of Measure

0.2.1 The big picture

The general idea of constructing measure is, in crude terms:

1. Find a collection of set that has a clear notion of size
2. Make the collection closed in finite set operations by generating an algebra on the collection.
3. Approximate arbitrary subsets using element in the algebra, hence get a notion of size on arbitrary subsets
4. Make the notion of size truly a measure by restricting to a special collection of subset.

More concretely the collection of set with clear notion of size need to have a certain structure called *elementary collection of set* (denoted by $\mathcal{E}$). *Elementary collection* has a special property: it can be promoted to an algebra $\mathcal{A}$ by disjoint union. Now the notion of size on $\mathcal{E}$ can be extended to the algebra $\mathcal{A}$, creating a so-called *pre-measure* which possess *countable additivity* on $\mathcal{A}$. Then the process of approximating arbitrary set using elementary class is called the construction of an *outer-measure*. During the process, we loose *countable additivity* as it is reduced to *countable subadditivity*. Finally we obtain a true *measure* by restricting the *outer-measure* on a special collection of subset called the *Caratheory σ -algebra*.

Observe how the notion of size ripens:

$$\begin{aligned} \text{Stage 1: } \boxed{|\cdot| : \mathcal{E} \rightarrow \mathbb{R}^+} &\longrightarrow \text{Stage 2: } \boxed{\mu_0 : \mathcal{A} \rightarrow \mathbb{R}^+} \\ &\longrightarrow \text{Stage 3: } \boxed{\mu^* : P(X) \rightarrow \mathbb{R}^+} \\ &\longrightarrow \text{Stage 4: } \boxed{\mu : \mathcal{S} \rightarrow \mathbb{R}^+} \end{aligned} \tag{12}$$

1. *Stage 1 to 2*: Expand the domain to ensures closedness and obtains *countable-additivity*; Lacks the ability to take arbitrary subset as input
2. *Stage 2 to 3*: Expand the domain to arbitrary subset but as a cost *countable additivity* deteriorates to *countable subadditivity*.
3. *Stage 3 to 4*: Shrink the domain; This is now a **complete measure**

0.2.2 Elementary collection and pre-measure

Definition 0.2.1 (Elementary Collection of set). An *elementary collection* of set is a collection of set $\mathcal{E} \in P(X)$ such that:

1. (*Contains null*) $\emptyset \in \mathcal{E}$
2. (*Closed under intersection*) If $E, F \in \mathcal{E}$ then $E \cap F \in \mathcal{E}$
3. (*Closed by disjoint union under complement*) If $F \in \mathcal{E}$ then exists disjoint subsets $E_j \in \mathcal{E}$ such that $F^c = \cup_j E_j$

0.2.3 Outer Measure

So how do we actually construct measures no matter what set we are working on?(Though in this book we are really interested in just $\mathbb{R}^n$). A general guideline is:

1. Find some *elementary set* $\mathcal{E}$ where premeasure is defined on the disjoint union of $\mathcal{E}$.
2. Define the measure of arbitrary subset $A \in P(X)$ as the the size of “smallest” cover of A using the elementary set:

$$\inf \left\{ \sum_{\nu=1}^{\infty} \rho(E_{\nu}) : E_{\nu} \in \mathcal{E}, A \subset \bigcup_{\nu=1}^{\infty} E_{\nu} \right\} \quad (13)$$

More generally we can define an *outer measure* as follows:

Definition 0.2.2 (Outer Measure). An *outer measure* on an non-empty set X is a function $\mu^* : P(X) \rightarrow [0, \infty]$ such that:

- $\mu^*(\emptyset) = 0$
- If $A \subset B$, then $\mu^*(A) \leq \mu^*(B)$
- $\mu^*(\cup_{\nu} A_{\nu}) \leq \sum_{\nu} \mu^*(A_{\nu})$

We note that property (2),(3) are both derived properties of measure. So we can think the definition of outer measure as a relaxation of measure: take out countable additivity and adds back two derived property. In addition, We note that (13) obeys Definition 0.2.2

Exercise

Show that (13) indeed defines an outer measure Definition 0.2.2.

How do we then obtain measure from outer measure? We note that what outer measure lacks is finite additivity, so we can further ask, is there a collection of subset $\mathcal{M} \subset P(X)$ such that μ^* is actually finitely additive? The answer is actually YES! However this seems rather arbitrary:

$$\mathcal{M} = \{A \subset X : \mu^*(F) = \mu^*(F - A) + \mu^*(A \cap F), \forall F \subset X\} \quad (14)$$

We call elements of $\mathcal{M}$ μ^* -**measurable**. In plain English, this saying A can “split measure” arbitrary $F \subset X$ by measuring the intersection and the complement separately and add them together.

Theorem 0.2.1 (Carathéodory’s Theorem). If μ^* is an outer measure of X then

1. the collection of μ^* -**measurable** sets:

$$\mathcal{M} = \{A \subset X : \mu^*(F) = \mu^*(A^c \cap F) + \mu^*(A \cap F), \forall F \subset X\} \quad (15)$$

is a σ -algebra.

2. $\mu^*|_{\mathcal{M}}$ is a **complete measure**

We first show a small lemma:

Lemma 0.2.2 ($\mathcal{M}$ is closed under finite union and). $\mathcal{M}\mu^*$ is **finitely additive** on $\mathcal{M}$
Let μ^* be an outer measure:

1. Suppose $A, B \in \mathcal{M}$ then

$$A \cup B \in \mathcal{M} \quad (16)$$

2. Suppose $A, B \in \mathcal{M}$, $A \cap B = \emptyset$

$$\mu^*(A \cup B) = \mu^*(A) + \mu^*(B) \quad (17)$$

proof of lemma. 1. (**$\mathcal{M}$ is closed under finite union**) It suffice to show that $A \cup B$ split measure every subset in X . That is, $\forall F \subset X$

$$\mu^*(F) = \mu^*(F \cap (A \cup B)) + \mu^*(F \cap (A \cup B)^c) \quad (18)$$

Since μ^* has subadditivity by definition, we only need to show:

$$\mu^*(F) \geq \underbrace{\mu^*(F \cap (A \cup B)) + \mu^*(F \cap (A \cup B)^c)}_I \quad (19)$$

where we denoted the quantity to be bounded as I . Now note that we can expand the two term again by subadditivity to get:

$$\mu^*(F \cap A \cap B) + \mu^*(F \cap A \cap B^c) + \mu^*(F \cap A^c \cap B) + \mu^*(F \cap A^c \cap B^c) \geq I \quad (20)$$

Now we note that $B \in \mathcal{M}$ hence B can split measure any set, including $F \cap A \subset X$ and $F \cap A^c \subset X$. Hence:

$$\mu^*(F \cap A) + \mu^*(F \cap A^c) \geq I \quad (21)$$

But $A \in \mathcal{M}$ therefore:

$$\mu^*(F) \geq I \quad (22)$$

2. (**μ^* is finitely additive on $\mathcal{M}$**) First use that $A \in \mathcal{M}$

$$\begin{aligned} \mu^*(A \cup B) &= \mu^*((A \cup B) \cap A) + \mu^*((A \cup B) \cap A^c) \\ &= \mu^*(A \cup (A \cap B)) + \mu^*(A^c \cap B) \end{aligned} \quad (23)$$

Now use disjointness to conclude: $A \cap B = \emptyset$, $A^c \cap B = B$ hence:

$$\mu^*(A \cup (A \cap B)) + \mu^*(A^c \cap B) = \mu^*(A) + \mu^*(B) \quad (24)$$

□

proof of Carathéodory. We only need to upgrade “finite” to “countable” in the previous argument.

□

0.3 Casestudy: Lebesgue-Stieltjes Measure

The most useful set that we would like to equip a measure with is, unarguably, the real numbers $\mathbb{R}$.

0.3.1 Starting Point: The half-intervals

Recall that we would start with an elementary collection that has a clear notion of size. We define the **h-intervals**:

$$H = \{(a, b] : a < b, a, b \in \mathbb{R}\} \quad (25)$$

We define a terminology here:

Definition 0.3.1 (Borel Sets). *Let X any metric space, the σ -algebra generated by the family of open sets in X is called the **Borel σ -algebra**, denoted by $\mathcal{B}_X$*

Exercise 2 (h-intervals generate $\mathcal{B}_{\mathbb{R}}$)

Check H is an elementary set and so it generates an algebra $\mathcal{A}$. Furthermore, show that the σ -algebra generated by $\mathcal{A}$ is $\mathcal{B}_{\mathbb{R}}$

Another terminology:

Definition 0.3.2 (Elementary Decomposition). *Suppose $A \in \mathcal{A}$, since $\mathcal{A} = \mathcal{M}(H)$ we can find finite disjoint h-intervals $\{I_j\}_{j=1}^N = \{(a_j, b_j]\}_{j=1}^N$ such that:*

$$A = \bigcup_{j=1}^N I_j \quad (26)$$

Then $\{I_j\}$ is an **elementary decomposition** of A .

We define size on H as follows:

Notion of Size on h-intervals

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and right-continuous, let $(a, b] \in H$, then:

$$|(a, b]| = F(b) - F(a) \quad (27)$$

Note when $F(x) = x$, we get the common notion of length of interval.

The next step is to extend this notion of size to $\mathcal{A} = \mathcal{M}(H)$.

0.3.2 Step 1: Construction of a premeasure

Now we have an algebra $\mathcal{A}$ where elements of the algebra $A \in \mathcal{A}$ has an elementary decomposition $A = \{I_j\}$ to h-intervals $I = (a_j, b_j]$, which is equipped with a notion of size $b_j - a_j$. A natural way to define the notion of size for A is hence:

$$\mu_0(A) = \sum_{j=1}^N |I_j| = \sum_{j=1}^N F(b_j) - F(a_j) \quad (28)$$

We claim that μ_0 is actually a *pre-measure* on $\mathcal{A}$.

Proposition 0.3.1. *Let $F : \mathbb{R} \rightarrow \mathbb{R}$ to be a right-continuous, increasing function. Define μ_0 as*

$$\begin{aligned} \mu_0 : \mathcal{A} &\rightarrow \mathbb{R} \\ \bigcup_{j=1}^N A &\mapsto \sum_{j=1}^N [F(b_j) - F(a_j)] \end{aligned} \quad (29)$$

where $\{I_j\} = \{(a_j, b_j]\}$ is an elementary decomposition. Then μ_0 is a premeasure on $\mathcal{A}$.

is a pre-measure. μ_0

We first need to show that μ_0 is actually well-defined¹. Then we illustrate:

1. Empty set has zero measure
2. Non-negativity
3. Finite additivity
4. Countable additivity where only (1), (2), (4) is truly required. But (3) is used in the proof of (4).

□

¹Note that in defining a notion of size μ_0 on $A \in \mathcal{A}$, we requires A to have a concrete representation $\cup_j I_j$. If the result is independent on how we choose this representation(i.e. different elementary decomposition $\{I_j\}$), then this notion of size is indeed fundamentally correct. Otherwise if μ_0 is dependent on how we choose representation, then it is not a credible notion of size. Such “representational invariance” is called *well-definedness* in mathematics.